Salient Object Detection

from Scratch

End-to-End Machine Learning / Deep Learning Project

Bledi Peci

ECSSD 1,000 images	**0.5890**	**0.6235**	**+5.9%**
Dataset	Baseline IoU	Improved IoU	IoU Gain

Cohort V | PyTorch | 128×128 px | 2026

Dataset & Pipeline

ECSSD – Extended Complex Scene Saliency Dataset

699	150	151
Train 69.9%	Val 15.0%	Test 15.1%

Preprocessing

- Resize ⟶ 128×128 px (bilinear/nearest)
- Normalise pixels ⟶ [0, 1]
- Binarise masks at threshold 0.5

Data Augmentation (training set only)

Horizontal flip 50%	Vertical flip 50%	Random rotation ±30° 50%
Random crop+resize 50%	Brightness jitter 50%	Contrast jitter 50%
Saturation jitter 50%	Gaussian noise 60%	Random erasing 70%

Model Architecture

BASELINE – BaselineSODNet

Stage	Ch	Out
Encoder 1	3⟶64	64×64
Encoder 2	64⟶128	32×32
Encoder 3	128⟶256	16×16
Encoder 4	256⟶512	8×8
Bottleneck + Drop2d	512⟶512	8×8
Decoder 4–1	512⟶32	128×128
Head Conv1×1 + Sigmoid	32⟶1	128×128

4,609,569 parameters

IMPROVED – ImprovedSODNet

1. **Skip connections (U-Net style)**

   Preserves spatial detail lost during pooling

2. **ASPP bottleneck (rates 1, 2, 4)**

   Multi-scale context at 8×8 feature map

3. **Attention gates on skips**

   Filters irrelevant background features

4. **Double conv encoder blocks**

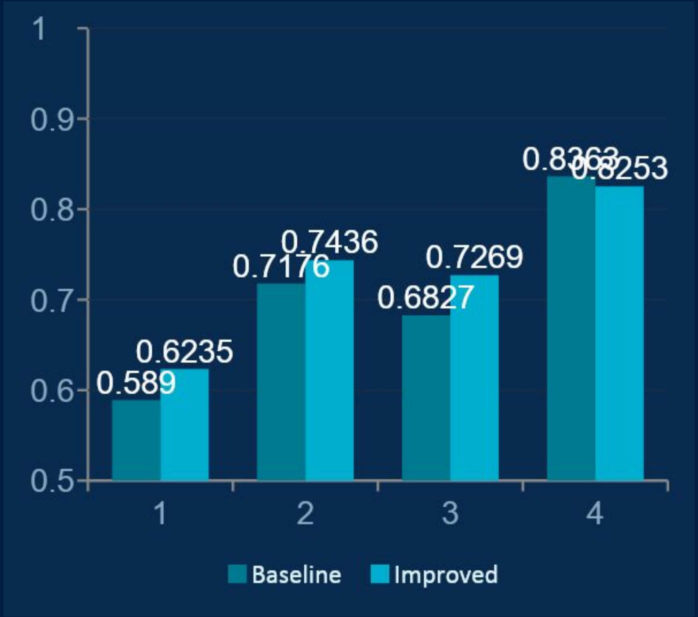
   Deeper per-stage representations

4,264,829 parameters

Experiments & Results

Test Set Evaluation (151 images, threshold = 0.5)

Metric	Baseline	Improved	Change
IoU	0.5890	**0.6235**	**+5.9%**
Precision	0.6827	**0.7269**	**+6.5%**
Recall	0.8363	**0.8253**	**−1.3%**
F1-Score	0.7176	**0.7436**	**+3.6%**
MAE	0.1658	**0.1501**	**−9.5%**



Loss: Focal-BCE (γ=2) + 1.0×(1−IoU) + 0.5×(1−SSIM) Optimizer: AdamW lr=5×10^{-4} Warmup 3 ep + Cosine decay Max 80 ep
Early stop patience=15

Demo & Key Learnings

⚡ Live Demo – Streamlit App

```
streamlit run app.py
```

- Upload any RGB image
- Adjust prediction threshold (slider)
- Saliency heatmap (HOT colour map)
- Overlay: input + red-tinted mask
- Inference time displayed
- ~15–30 ms per image on CPU

Key Learnings

Skip connections

Single most impactful change — recovers spatial detail discarded by MaxPool

Combined loss

Focal-BCE + IoU + SSIM produces sharper, more coherent masks than BCE alone

Dataset size

Primary bottleneck; 699 images demands heavy augmentation & regularisation

AdamW

Outperforms Adam on small datasets due to correct weight-decay scaling

LR scheduling

Warmup + cosine annealing eliminates IoU oscillations vs ReduceLROnPlateau